

COMP 2121

DISCRETE
MATHEMATICS

Name:

Student ID:

Quiz 4

Set: A B C D (circle one)

- Time allowed: 20 min.

1. (2 marks) Given

$$\sum_{k=45}^{101} (2a_k) = 150 \quad \text{and} \quad \sum_{i=47}^{54} a_{i-2} + \sum_{j=53}^{102} (a_{j-1} + 2) = 160$$

find a_{52} .

$$\sum_{k=45}^{101} (2a_k) = 150$$

$$2 \sum_{k=45}^{101} a_k = 150$$

$$\sum_{k=45}^{101} a_k = \frac{150}{2}$$

$$\sum_{k=45}^{101} a_k = 75$$

$$\sum_{i=47}^{54} a_{i-2} + \sum_{j=53}^{102} (a_{j-1} + 2) = 160$$

$$\sum_{k=45}^{52} a_k + \sum_{j=53}^{102} a_{j-1} + \sum_{j=53}^{102} (2) = 160$$

$$\sum_{k=45}^{52} a_k + \sum_{k=52}^{101} a_k + (102-53+1)(2) = 160$$

a_{52} is here twice

$$\sum_{k=45}^{101} a_k + a_{52} = 160 - 50(2)$$

$$75 + a_{52} = 60$$

$$a_{52} = 60 - 75$$

$$a_{52} = -15$$

2. (3 marks) Use the formula $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ to find the value of the counter after the following programming segment is executed

```

counter = 300;
for (i = 4; i <= n; i++){
    counter = counter + 2i
    for (j = i; j <= n+6; j++){
        counter = counter + 4
    }
}
//assume n>5

```

*j loop runs for all values of i,
no adjustment needed.*

Simplify your answer as far as possible.

$$\text{Counter} = 300 + \sum_{i=4}^n \left[2i + \sum_{j=i}^{n+6} (4) \right] \quad \text{const.}$$

$$= 300 + \sum_{i=4}^n \left[2i + (n+6-i+1)(4) \right] = 300 + \sum_{i=4}^n \left[2i + (n+7-i)(4) \right]$$

$$= 300 + \sum_{i=4}^n \left[2i + 4n + 28 - 4i \right] = 300 + \sum_{i=4}^n \left[\underbrace{4n + 28}_{\text{const}} - 2i \right]$$

$$= 300 + \sum_{i=4}^n (4n + 28) - \sum_{i=4}^n 2i = 300 + (n-4+1)(4n+28) - 2 \sum_{i=4}^n i$$

$$= 300 + (n-3)(4n+28) - 2 \left(\sum_{i=1}^n i - \sum_{i=1}^3 i \right)$$

$$= 300 + 4n^2 + 28n - 12n - 84 - 2 \left(\frac{n(n+1)}{2} - \frac{3(4)}{2} \right)$$

$$= 300 + 4n^2 + 16n - 84 - n^2 - n + 12 = \boxed{3n^2 + 15n + 228}$$